

Karachi Food Delivery Time Analysis — Auto Analysis

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Executive Summary

This report summarizes the dataset quality, exploratory findings, and model performance for project "Karachi Food Delivery Time Analysis".

Dataset Overview

File	karachi_food_delivery_dataset.csv
Rows	2500
Columns	18
Missing Values	0
Duplicate Rows	0
Cleaned	True

Modeling & Evaluation

Model Type	gradient_boosting
Problem Type	classification
Target Column	late_delivery

Metrics

Metric	Value
accuracy	0.91
precision	0.909
recall	0.91
f1	0.9068
roc_auc	0.9591
confusion_matrix	{'labels': ['No', 'Yes'], 'matrix': [[364, 11], [34, 91]]}

class_distribution	{'No': 375, 'Yes': 125}
cv_scores	[0.9110850333995306, 0.8654646324549238, 0.89543986352497, 0.8829131652661064, 0.9036682998196727]
cv_mean	0.8917
baseline	{'baseline_score': 0.6429, 'baseline_metric': 'f1', 'model_score': 0.9068, 'improvement': 0.2639, 'cv_std': 0.0161, 'clearly_beats_baseline': True}
leakage_warnings	[{'column': 'delivery_time_min', 'correlation': 0.9984, 'reason': "Together with 'promised_time_min', this column almost exactly reproduces the target (99.8% agreement via a simple comparison) — the model could 'cheat' by combining the two instead of learning real patterns."}, {'column': 'promised_time_min', 'correlation': 0.9984, 'reason': "Together with 'delivery_time_min', this column almost exactly reproduces the target (99.8% agreement via a simple comparison) — the model could 'cheat' by combining the two instead of learning real patterns."}]
post_outcome_excluded	[{'column': 'tip_pkr', 'reason': "'tip_pkr' often represents information only available AFTER the outcome occurs (e.g. a rating or tip given once an order is complete), and shows a real (34% correlated) relationship with the target here. Review whether this would actually be known at prediction time — if not, exclude it below."}, {'column': 'customer_rating', 'reason': "'customer_rating' often represents information only available AFTER the outcome occurs (e.g. a rating or tip given once an order is complete), and shows a real (46% correlated) relationship with the target here. Review whether this would actually be known at prediction time — if not, exclude it below."}]
post_outcome_included	[]

Feature Importance

Feature	Importance
distance_km	0.3107
prep_time_min	0.2022
traffic_level_Jam	0.1343
weather_Rainy	0.0711
vehicle_type_Bicycle	0.0642
traffic_level_Low	0.0604
traffic_level_High	0.0339

order_value_pkr	0.027
rider_experience_months	0.0215
items_count	0.0166
vehicle_type_Motorbike	0.0148
rider_rating	0.0126
traffic_level_Medium	0.0098
weather_Clear	0.0086
order_time_dayofweek	0.0032
area_Clifton	0.0022
order_time_month	0.0016
area_Gulshan-e-Iqbal	0.0012
area_DHA	0.0008
cuisine_Pizza	0.0007

AI Business Insight AI-generated insight

AI service is currently unavailable. This report contains verified analytical results only; AI-generated narrative could not be produced.

Limitations

Model predictions are estimates based on historical data and may contain uncertainty. AI-generated text explains verified analytical results but does not independently calculate metrics. Always validate significant business decisions with domain expertise.